

CSE 2600: Introduction to Data Science and Engineering

Assignment 4

Instructions:

- You must support your answers to receive credit.
 - Answers can be typed or handwritten, and should be readable.
 - Submit the assignment in one file via PDF on HuskyCT.
1. (10 points) Suppose an exam is to be given to students at a local college. To test its validity, the exam is administered to a sample of 10 students, and the scores are:

65, 70, 68, 73, 75, 80, 85, 78, 90, 100.

We will estimate the 95% confidence interval for the mean of the population (i.e., the expected mean for a large number of students) using two different methods:

- (a) Calculate the sample mean $\bar{x}$ and standard deviation s of this set. To calculate the confidence interval $[\bar{x} - r, \bar{x} + r]$, use the formula

$$r = t \cdot \frac{s}{\sqrt{n}},$$

where n is the number of records in the sample, and $t = 2.262$ (the critical value corresponding to the 95% confidence level and degrees of freedom $n - 1 = 9$).

- (b) Construct 40 bootstrapped samples of the data. (You might want to create a function in Python to do this quickly.) Calculate the mean of each sample, and use this set of means to estimate the 95% confidence interval for the mean of the population.
 - (c) Which method would you choose to estimate the confidence interval for this use case. Explain why you believe your chosen method is more reliable?
2. (10 points) Using the same 10 exam scores as in question 1, we will use **cross-validation** to estimate the prediction error of an exam score on new (holdout) data.
 - (a) Randomly assign each score into one of 5 subsets, each containing only two scores. Include a snippet of your code along with the final outcome.
 - (b) For each of the five subsets created, we have 8 associated “training” scores and 2 “validation” scores (i.e., if one of your subsets is [65, 100], then the associated training set is [70, 68, 73, 75, 80, 85, 78, 90]). For each of the 5 training sets, calculate the average score (the simple mean of the 8 elements).
 - (c) If we treat each of the 5 calculated average scores as a prediction, what is the mean absolute error computed on each associated validation set (each containing the 2 remaining elements)?
 - (d) Compute the average mean absolute error across the 5 cross-validation folds.
 - (e) Suppose we conduct this exam again, and for simplicity let’s assume we get exactly the same scores. If we had used the average exam score to predict the “new” exam scores, what would the mean absolute error be on this “new” set?

3. (30 points) Adapted from ISLP 8.4.9.

This problem involves the `OJ` data, which is part of the `ISLP` package. We will use classification trees to predict the value of `Purchase`.

- (a) Create a training set contain a random sample of 800 observations, and a test set containing the remaining observations. You will also need to convert the `Store7` column to numeric. Provide a screenshot of the code used for this data engineering.
- (b) Fit a single tree to the training data, with `Purchase` as the response and the other variables as predictors. Use 6 as the maximum number of leaf nodes. Produce a confusion matrix and state the total misclassification error on training data.
- (c) Create a plot of the tree and interpret the results.
- (d) Predict the response on the test data, and produce a confusion matrix. State the total misclassification error on test data.
- (e) Fit a new tree to the data, this time using 8 for the maximum number of leaf nodes. How does the test error change?
- (f) Is this new model in part (e) overfit? Why or why not?
- (g) Use cross-validation to determine the optimal number of leaf nodes. When you train a model using this number of leaf nodes, does the total misclassification error on test decrease?
- (h) Create a random forest model to predict `Purchase`. What are your initial hyperparameters? What is the total misclassification rate on the test set?
- (i) Iterate on your random forest model to search for optimal hyperparameters. What are your chosen final hyperparameters? What is the total misclassification rate?
- (j) Construct a lift chart for your final random forest model on test data. Use it to estimate the total segmentation of your model.
- (k) Create a gradient boosting model to predict `Purchase`. What are your initial hyperparameters? What is the total misclassification rate on the test set?
- (l) Iterate on your gradient boosting model to search for optimal hyperparameters. What are your chosen final hyperparameters? What is the total misclassification rate?
- (m) Construct a dual lift chart which compares your best random forest model to your best gradient boosting model. Use it to explain which model better segments the data.